PCBC H Parametric & Sequences Test (Practice B) **Name**_____ _____/27

SHOW ALL WORK FOR FULL CREDIT. ***Calculator*** **You have the entire period.**

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test.

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_____ _____

Signature

Solve the following application problem. Report answers to the thousandths. Include units in your answers. (4 points)

7. A softball is thrown from a height of 5 feet at an angle of 40° with an initial speed of 70 ft/s.

a) Write a set of parametric equations for the motion of the softball. (Recall: $x = (v\cos\theta)t,\ \ y = (v\sin\theta)t - 16t^2 + h$)

b) Determine the maximum height reached by the softball.

c) Determine how long the softball is in the air.

d) Determine how far the softball traveled in the air.

The given pattern continues. Write down the nth term of the sequence $\{a_n\}$ suggested by the pattern. (3 points)

8. $\frac{1}{2}, -\frac{4}{4}, \frac{9}{8}, -\frac{16}{16}, \cdots$

Express the sum using summation notation. (3 points)

9. $\frac{1}{2} - \frac{2}{3} + \frac{3}{4} - \cdots - \frac{10}{11}$

Find each sum. Show the properties and formulas you use, then use your calculator. (#10 and #11 are 3 points each, #12 is 4 points)

10. $\sum_{k=1}^{24} (10 - 3k)$

11. $\sum_{k=1}^{18} 2k(k-1)$

12. $\sum_{k=6}^{25} 5k^2$

Expand using the Binomial Theorem. (#13 is 3 points, #14 is 4 points)

13. $(x-2)^8$

14. $(3x-2)^4$

Extra practice: more sequence and summation questions

Not part of the 27 points. Worked in the key.

Write out the sum.

E1. $\displaystyle\sum_{k=1}^{n} (-1)^k \frac{k+1}{k}$

The given pattern continues. Write down the nth term of the sequence $\{a_n\}$ suggested by the pattern.

E2. $-3,\ 6,\ -9,\ 12,\ \ldots$

Answer Key

Work on the left, the reason beside it. Boxed = final answer.

Day 2: Calculator

7. Softball: $h = 5$ ft, $\theta = 40°$, $v = 70$ ft/s. Calculator in **degree** mode, Graph in **Parametric**.

a) $x = (70\cos 40°)t$ — *plug v and θ into the given formula; degree symbol required*

$y = (70\sin 40°)t - 16t^2 + 5$ — *+h: starts at 5 ft*

$x_1(t) = 70\cos(40°)\cdot t,\ \ y_1(t) = 70\sin(40°)\cdot t - 16t^2 + 5$ — *what you type in*

$0 \le t \le 3$, tstep 0.01; x : 0 to 170, y : 0 to 42 — *graph doesn't come back down? raise tmax*

b) $\boxed{y \approx 36.634 \text{ ft}}$ — *trace to the top ($t \approx 1.406$); not Analyze, Maximum*

c) $\boxed{t \approx 2.919 \text{ s}}$ — *trace to $y \approx 0$; trace step 0.01, then 0.001*

d) $\boxed{x \approx 156.539 \text{ ft}}$ — *same point, read x*

Units: feet (the speed was ft/s).

8. $\frac{1}{\frac{1}{2}}, -\frac{2}{\frac{4}{4}}, \frac{3}{\frac{9}{8}}, -\frac{4}{\frac{16}{16}}, \ldots$

top: n^2, bottom: 2^n — *write 1 2 3 4 above; top and bottom separately*

$+,-,+,- \Rightarrow (-1)^{n+1}$ — *first term positive*

$\boxed{a_n = (-1)^{n+1}\frac{n^2}{2^n}}$ — *check $n = 1$: $\frac{1}{2}$*

9. $\frac{1}{2} - \frac{2}{3} + \frac{3}{4} - \cdots - \frac{10}{11}$

top k, bottom $k+1$ — *top counts 1 to 10*

$+,-,+ \Rightarrow (-1)^{k+1}$ — *first term positive; $k = 10$ is negative, matches*

$\boxed{\sum_{k=1}^{10} (-1)^{k+1}\frac{k}{k+1}}$

10.

$10\sum_{k=1}^{24} 1 - 3\sum_{k=1}^{24} k$ — *(2), (3) split; (1) constants out*

$10(24) - 3\left[\frac{24(25)}{2}\right]$ — *(5), (6) with $n = 24$; then the calculator*

$\boxed{-660}$

11. $2k(k-1) = 2k^2 - 2k$ first

$2\sum_{k=1}^{18} k^2 - 2\sum_{k=1}^{18} k$ — *(2), (3) split; (1) constants out*

$2\left[\frac{18(19)(37)}{6}\right] - 2\left[\frac{18(19)}{2}\right]$ — *(7), (6) with $n = 18$; then the calculator*

$\boxed{3876}$

12.

$5\sum_{k=6}^{25} k^2$ — *(2), (3) split; (1) constants out*

$5\left[\sum_{k=1}^{25} k^2 - \sum_{k=1}^{5} k^2\right]$ — *(4): starts at 6, so $j=5$*

$5\left[\frac{25(26)(51)}{6} - \frac{5(6)(11)}{6}\right]$ — *(7); then the calculator*

$\boxed{27350}$

13. $(x-2)^8$: $n=8$, row 1 8 28 56 70 56 28 8 1. The sign goes with its term.

$1x^8(-2)^0 + 8x^7(-2)^1 + 28x^6(-2)^2 + 56x^5(-2)^3$ — *blanks first: first term's powers count down, second term's count up*

$+70x^4(-2)^4 + 56x^3(-2)^5 + 28x^2(-2)^6 + 8x^1(-2)^7$

$+1x^0(-2)^8$

$\boxed{x^8 - 16x^7 + 112x^6 - 448x^5 + 1120x^4 - 1792x^3 + 1792x^2 - 1024x + 256}$

Second term negative: odd powers of -2 are negative, so signs alternate.

14. $(3x-2)^4$: $n=4$, row 1 4 6 4 1. The sign goes with its term.

$1(3x)^4(-2)^0 + 4(3x)^3(-2)^1 + 6(3x)^2(-2)^2 + 4(3x)^1(-2)^3$ — *blanks first: first term's powers count down, second term's count up*

$+1(3x)^0(-2)^4$

$(3x)^4 = 81x^4, \ldots$ — *the whole first term gets the power*

$\boxed{81x^4 - 216x^3 + 216x^2 - 96x + 16}$

Second term negative: odd powers of -2 are negative, so signs alternate.

Extra practice (not scored)

E1. $\sum_{k=1}^{n} (-1)^k \frac{k+1}{k}$

$(-1)^1\frac{2}{1} + (-1)^2\frac{3}{2} + (-1)^3\frac{4}{3} + \cdots + (-1)^n\frac{n+1}{n}$ — *$k=1$ is negative*

$\boxed{-2 + \frac{3}{2} - \frac{4}{3} + \cdots + (-1)^n\frac{n+1}{n}}$ — *don't add*

E2. $\overset{1}{-3}, \overset{2}{6}, \overset{3}{-9}, \overset{4}{12}, \ldots$

top: $3n$, bottom: 1 — *write 1 2 3 4 above; top and bottom separately*

$-,+,-,+ \Rightarrow (-1)^n$ — *first term negative*

$\boxed{a_n = (-1)^n 3n}$